

GENIFY

LECTURE 07

30-July-2025

TOPICS COVERED:

RIS

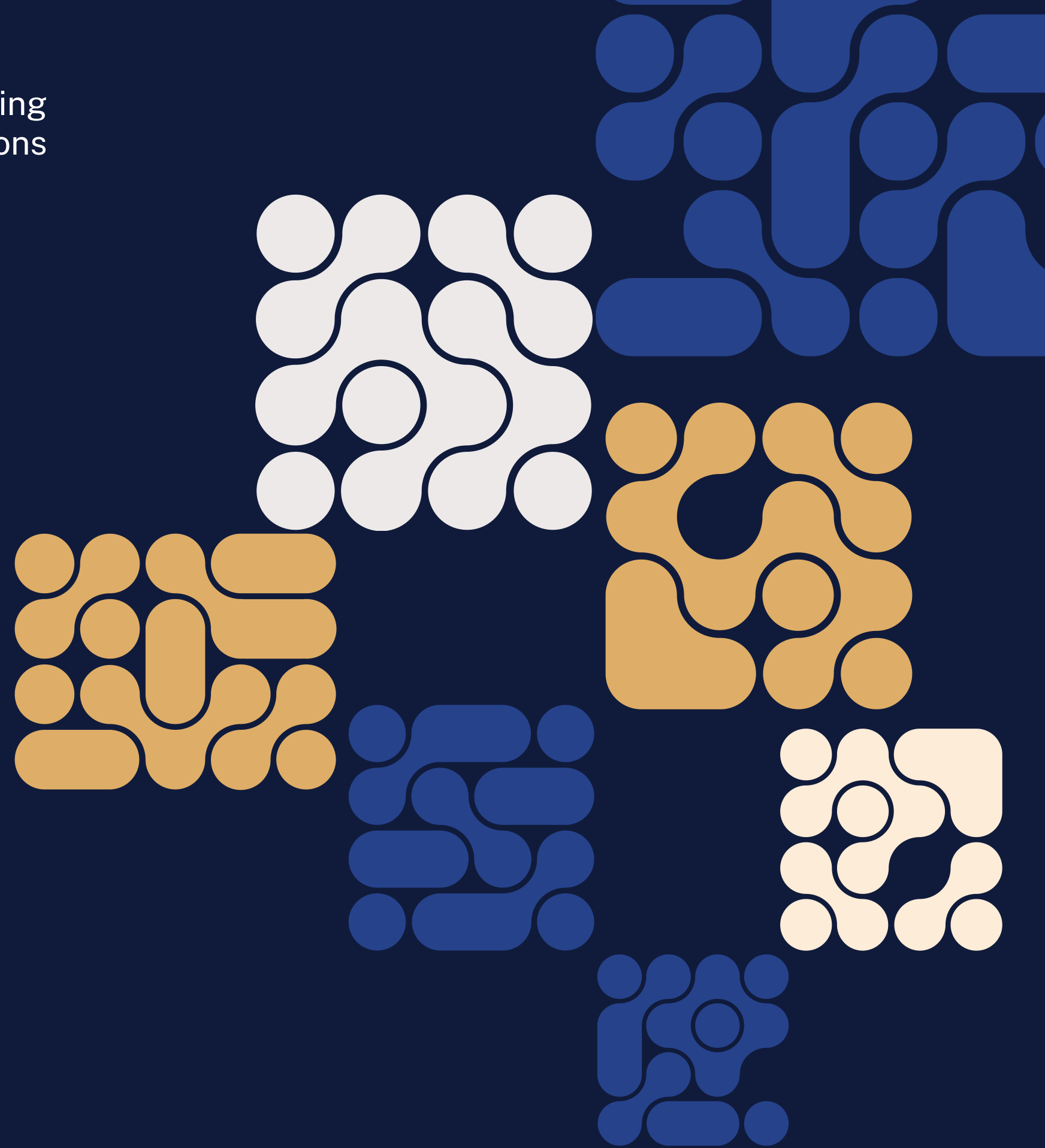
- RISC-V SINGLE CYCLE CORE DEEP DIVE

Presenter

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RECAP QUIZ TIME

ASSIGNMENT DISCUSSION

1. Multiplication Table Generator

Write a RISC-V assembly program that generates and prints the multiplication table of a given number (e.g., 7) from 1 to 10. Store the result of each multiplication in a register and simulate printing using comments or storing in memory.

2. Fibonacci Sequence in Two Registers

Write a RISC-V assembly program that calculates the Fibonacci sequence. Use 2 registers to Store initial Fibonacci sequence values (i.e 0 and 1) and then run the sequence until a given number N (e.g 6)

Example:

If N = 6, then at the end of program registers should contain 55 and 89 (11th and 12th Fibonacci numbers, 6 iterations done).

3. Sum of First N Natural Numbers

Write a RISC-V program that computes the sum of the first N natural numbers. Store the final result in a register.

4. Count Even Numbers in a Range

Write a RISC-V program that counts how many even numbers are present between 1 and N (inclusive), and stores the result in a register.

5. Reverse a Number

Given a positive integer in a register (e.g., 1234), write a RISC-V program that reverses its digits (4321) and stores the result in another register.

RISC-V Exercises

6. Find the Maximum of an Array

Given an array of N integers stored in memory, write a RISC-V assembly program to find the maximum value and store it in a register.

7. Factorial of a Number

Write a RISC-V program to compute the factorial of a number N and store the result in a register.

8. Compare Two Numbers

Write a RISC-V assembly program that compares two numbers stored in registers and sets a flag (1 or 0) in another register based on whether the first number is greater than the second.

9. Swap Two Registers Without Using a Third Register

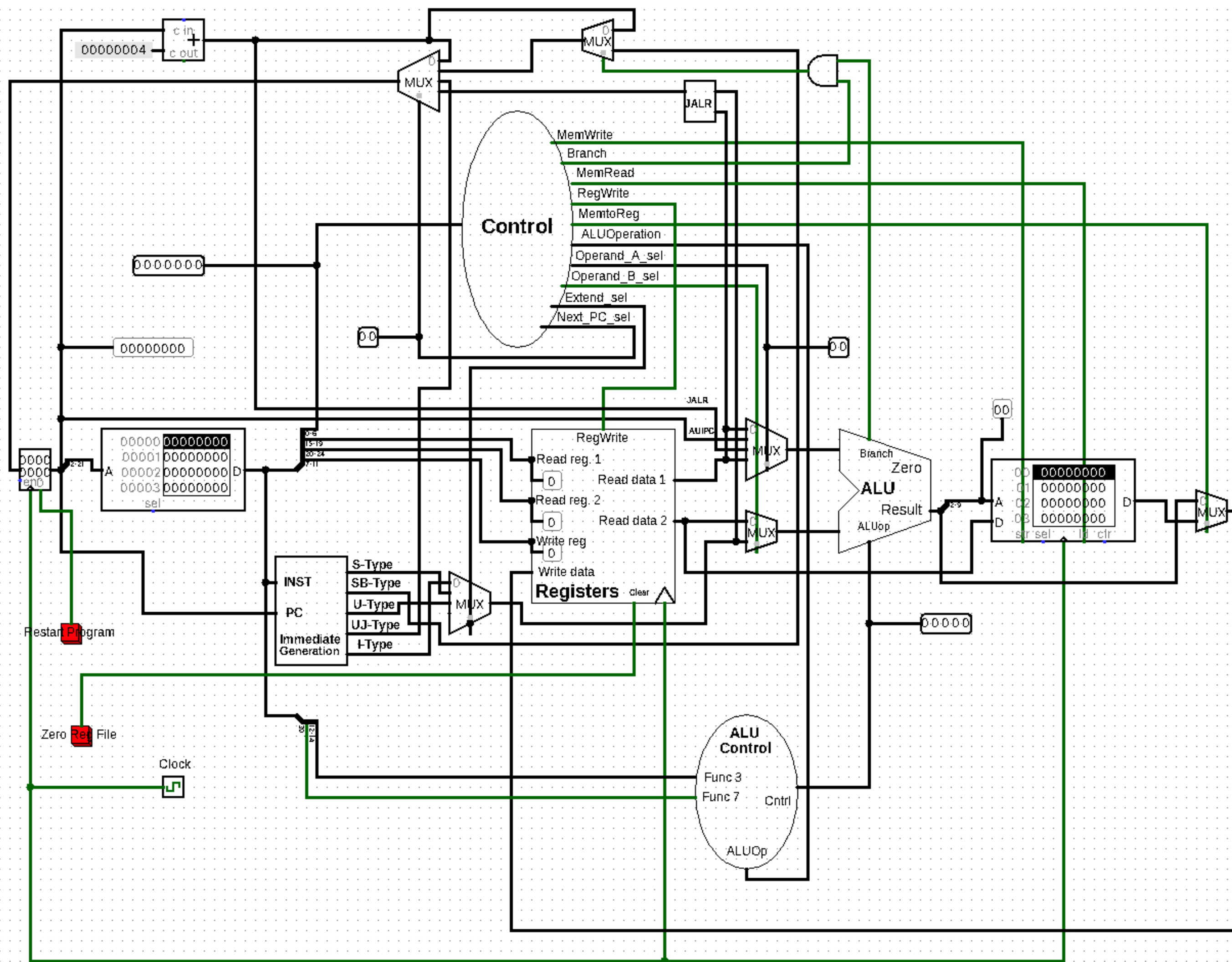
Write a RISC-V program that swaps the values of two registers without using a third temporary register.

10. Print All Odd Numbers till N

Write a RISC-V program to print (or store) all odd numbers from 1 to N using a loop.



RISC-V SINGLE CYCLE CORE



RISC-V Green Card

Green Card Resource: https://www.eecs.yorku.ca/course_archive/2019-20/F/2021S/REF/RISCV-GreenCard.pdf

Codec: <https://luplab.gitlab.io/rvcodecjs/>

Logisim Datapath: https://github.com/merledu/genify/blob/main/lecture6/risc5_single_cycle.circ